Sanjaikumar Balasubramaniyan

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Summary

AI/ML Engineer in a client-facing consulting role, shipping production LLM and transformer-based systems on Lakehouse infrastructure. Experience building ML pipelines over millions of records, cutting inference latency up to 98% on GPUs, and deploying conversational AI used by business teams. Strong in PyTorch, Databricks, RAG, and LLM evaluation.

Education

Master of Science in Data Science GPA: 3.51/4
Indiana University – Bloomington, IN, USA *August 2024 – May 2026*

Coursework: Foundations of Applied Data Science (ML & LLMs), Data Mining, Applied Algorithms, Statistics, Advanced Database Concepts, Engineering Cloud Computing

Bachelor of Technology in Artificial Intelligence and Data Science
SRM Institute of Science and Technology (SRMIST) – Chennai, India *October 2020 – April 2024*

Work Experience

AI/ML Engineer June 2026 – Present
Agilisium Consulting *Atlanta, GA, USA*

- Embedded with a global pharmaceutical client, architecting a production conversational AI system (text-to-SQL on Databricks Genie, in Microsoft Teams) giving field sales teams governed access to a national-scale claims dataset
- Validating a transformer-based prediction model head-to-head against an independently built model, designing the benchmarking methodology that decides which moves to production
- Contributing to a Lakehouse architecture (Unity Catalog, MLflow) processing healthcare claims data at scale
- Driving governance, access-control, and rollout design decisions with client architecture and infra teams

Machine Learning Researcher Jan 2025 – June 2026
Indiana University *Bloomington, IN, USA*

- Designed a taxonomy-grounded LLM classification framework, reducing off-taxonomy hallucinations to near zero across 26 categories over 3 refinement cycles
- Cut batch inference runtime 98% (60+ min → under 1 min) by moving RoBERTa and LLM pipelines to GPU HPC
- Built an uncertainty-aware multi-label predictor with calibrated probabilities for cases single-label baselines missed
- Built a reproducible benchmarking pipeline over thousands of documents, standardizing evaluation across iterations

Founding AI Engineer Intern July 2025 – Nov 2025
riAI Capital LLC *Reno, NV, USA*

- Architected and deployed a RAG-based LLM system over 5M+ financial records, cutting research turnaround time 40% via optimized retrieval and ranking logic
- Built evaluation and benchmarking infrastructure from scratch, enforcing response consistency and replacing ad-hoc manual review in production
- As sole engineer, owned backend, real-time inference, monitoring, and KPIs for a compliance-sensitive system

Projects

LLM Fine-Tuning & Evaluation Framework | *PyTorch, Hugging Face, LoRA, Weights & Biases*

- Fine-tuned an LLM with **LoRA**, cutting **trainable parameters 90%** at equal task performance; tracked **BLEU**, **ROUGE**, and hallucination rates across iterations

Semantic Search Engine with RAG Pipeline | *Python, LangChain, ChromaDB, Gemini API, Streamlit*

- Built an end-to-end **RAG pipeline** integrating **LangChain**, **ChromaDB**, and **Gemini embeddings** for semantic search and QA over PDFs, streamed via Streamlit

Technical Skills & Certifications

ML & AI: Transformers, LoRA Fine-tuning, RAG, Text-to-SQL, Prompt Engineering, Embeddings, PyTorch, TensorFlow
Scale & Infra: GPU Profiling, Inference Optimization, Databricks, Unity Catalog, MLflow, Azure Data Factory, Benchmarking
Core ML: Model Training & Evaluation, Ranking/Recommendation Systems, Information Retrieval, Feature Engineering
Programming: Python, C/C++, R, SQL **Tools**: Git, Docker, AWS, Apache Airflow, Weights & Biases, Jira, Scrum
Certification: Databricks Certified Machine Learning Associate